

Elastic Fields of a Screw Dislocation

1. Displacement Field (Screw Dislocation)

For a screw dislocation with Burgers vector b along the z -axis:

$$u_z = \frac{b}{2\pi} \tan^{-1}\left(\frac{y}{x}\right), \quad u_x = 0, \quad u_y = 0.$$

Let

$$\theta = \tan^{-1}\left(\frac{y}{x}\right), \quad r = \sqrt{x^2 + y^2}.$$

2. Strain Field Derivations

All normal strains vanish:

$$\varepsilon_{xx} = 0, \quad \varepsilon_{yy} = 0, \quad \varepsilon_{zz} = 0.$$

Shear strains are nonzero only in components involving z :

$$\varepsilon_{xz} = \frac{1}{2} \frac{\partial u_z}{\partial x} = -\frac{b}{4\pi r} \sin \theta,$$

$$\varepsilon_{yz} = \frac{1}{2} \frac{\partial u_z}{\partial y} = \frac{b}{4\pi r} \cos \theta.$$

$$\varepsilon_{xy} = 0.$$

Trace:

$$\varepsilon_{kk} = 0.$$

3. Hooke's Law (Isotropic Elasticity)

Since the only nonzero strains are shear strains and the trace is zero:

$$\sigma_{xz} = 2G\varepsilon_{xz}, \quad \sigma_{yz} = 2G\varepsilon_{yz}, \quad \sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = 0.$$

4. Stress Field Derivations

$$\sigma_{xz}$$

$$\sigma_{xz} = 2G\varepsilon_{xz} = 2G \left(-\frac{b}{4\pi r} \sin \theta \right),$$

$$\sigma_{xz} = -\frac{Gb}{2\pi r} \sin \theta$$

$$\sigma_{yz}$$

$$\sigma_{yz} = 2G\varepsilon_{yz} = 2G \left(\frac{b}{4\pi r} \cos \theta \right),$$

$$\sigma_{yz} = \frac{Gb}{2\pi r} \cos \theta$$

5. Final Classical Stress Fields (Screw Dislocation)

$$\sigma_{xz} = -\frac{Gb}{2\pi r} \sin \theta$$

$$\sigma_{yz} = \frac{Gb}{2\pi r} \cos \theta$$

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma_{xy} = 0.$$

These are the standard textbook stresses of a screw dislocation.